Large Language Models
Homework 1

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1 N-Gram, RNN, and LSTM Models for Language Identification

1.1 Setup

The course-provided splits are used without modification: 1,120 training, 140 validation, and 140 test texts per language (5,600 / 700 / 700 in total). A shared character vocabulary of size $|V| = 129$ is built from `train.csv` only, including the special tokens `<PAD>`, `<UNK>`, `<BOS>`, and `<EOS>`. Characters that never occur in the training data are mapped to `<UNK>`. All models use the same vocabulary and the same special-token policy: `<BOS>` and `<PAD>` are never prediction targets, while `<EOS>` is. Every text is assigned to the language whose model gives the lowest perplexity,

$$\hat{l} = \arg\min_{l \in \mathcal{L}} \mathrm{PPL}(x \mid l),$$

and all validation and test perplexities are corpus-level: the total negative log-likelihood under the ground-truth language model divided by the total number of prediction targets. Per-text perplexities are never averaged.

1.2 Character-Level Bigram and Trigram Models

For each language, bigram and trigram counts are collected from its training texts and smoothed with Add-α ($\alpha = 1$):

$$p(c_t \mid c_{t-1}, l) = \frac{C_l(c_{t-1}, c_t) + \alpha}{C_l(c_{t-1}) + \alpha|V|}, \qquad p(c_t \mid c_{t-2}, c_{t-1}, l) = \frac{C_l(c_{t-2}, c_{t-1}, c_t) + \alpha}{C_l(c_{t-2}, c_{t-1}) + \alpha|V|}.$$

Each text is left-padded with $n-1$ `<BOS>` tokens. As a result, the first character is also predicted, and both models score exactly the same targets: every character plus `<EOS>`.

Table 1: Results of the n-gram models.

Model	Val PPL	Test PPL	Accuracy	Macro-F1	Build time (s)	Inference time (s)
Bigram	11.4373	11.3264	0.9729	0.9727	1.913	0.166
Trigram	**9.7008**	**9.4022**	**0.9871**	**0.9871**	2.120	0.184

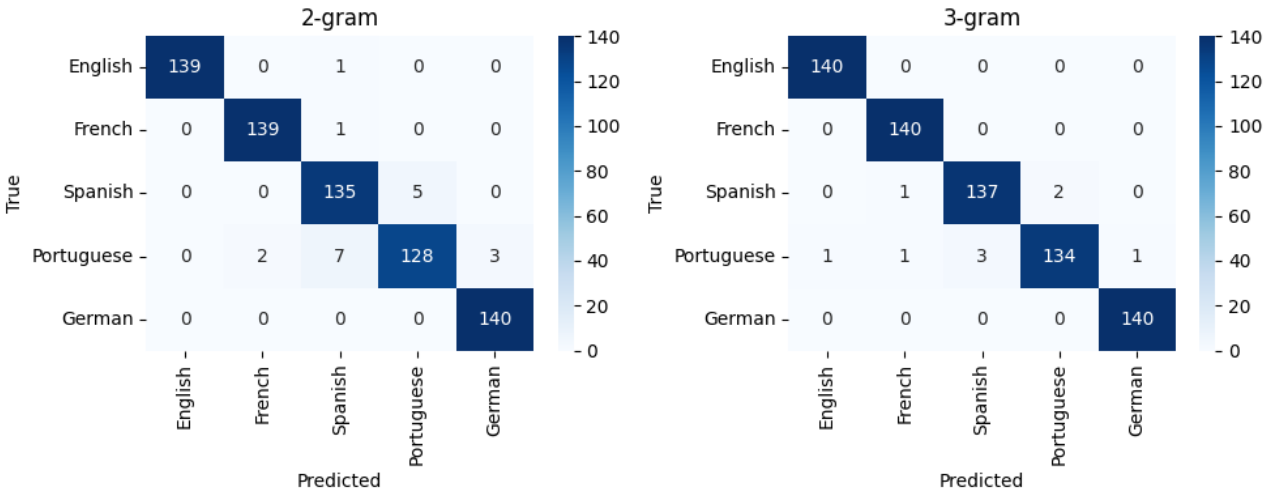


Figure 1: Test confusion matrices of the bigram (left) and trigram (right) models.

Results. The trigram model is better than the bigram model on every metric. Its validation perplexity is 9.70 instead of 11.44, and its test accuracy is 0.9871 instead of 0.9729. With two characters of context, the trigram model can capture language-specific patterns such as *-ção* (Portuguese) versus *-ción* (Spanish), or *sch* and *th*.

Cost. Both models are built in about 2 s on a CPU and need very little memory. The peak memory during construction is 0.42 MB for the bigram and 2.64 MB for the trigram. The trigram uses more memory because the number of possible contexts grows from $|V|$ to $|V|^2$. Classifying all 700 test texts takes less than 0.2 s.

Error analysis. As Figure 1 shows, almost all errors involve Portuguese or Spanish. The bigram model misclassifies 12 Portuguese texts (7 as Spanish) and 5 Spanish texts (as Portuguese). The trigram model reduces this to 6 Portuguese and 3 Spanish errors. Representative errors are mostly very short or contain little language-specific evidence:

- "`4.800 --`" (Portuguese, 8 characters), predicted as German by both n-gram models. It contains no letters at all.
- "*Decidimos,*" (Portuguese), predicted as Spanish. The word is spelled identically in both languages.
- "*to protect us*" (English), predicted as Spanish, and "*un monde de murs,*" (French), predicted as Spanish.
- "*Ándale! Ándale!*" in the Portuguese test set, predicted as German. This is a Spanish expression quoted inside a Portuguese talk.

1.3 Character-Level RNN Language Models

Five character-level language models are trained, one per language, using the suggested settings: embedding dimension 128, hidden dimension 128, 2 layers, Adam with learning rate 0.001, 10 epochs, batch size 32, and cross-entropy loss. The input is `<BOS>` followed by the characters of the text. The target is the next character, ending with `<EOS>`, and padded positions are ignored in the loss. For each language, the checkpoint with the lowest validation perplexity is kept.

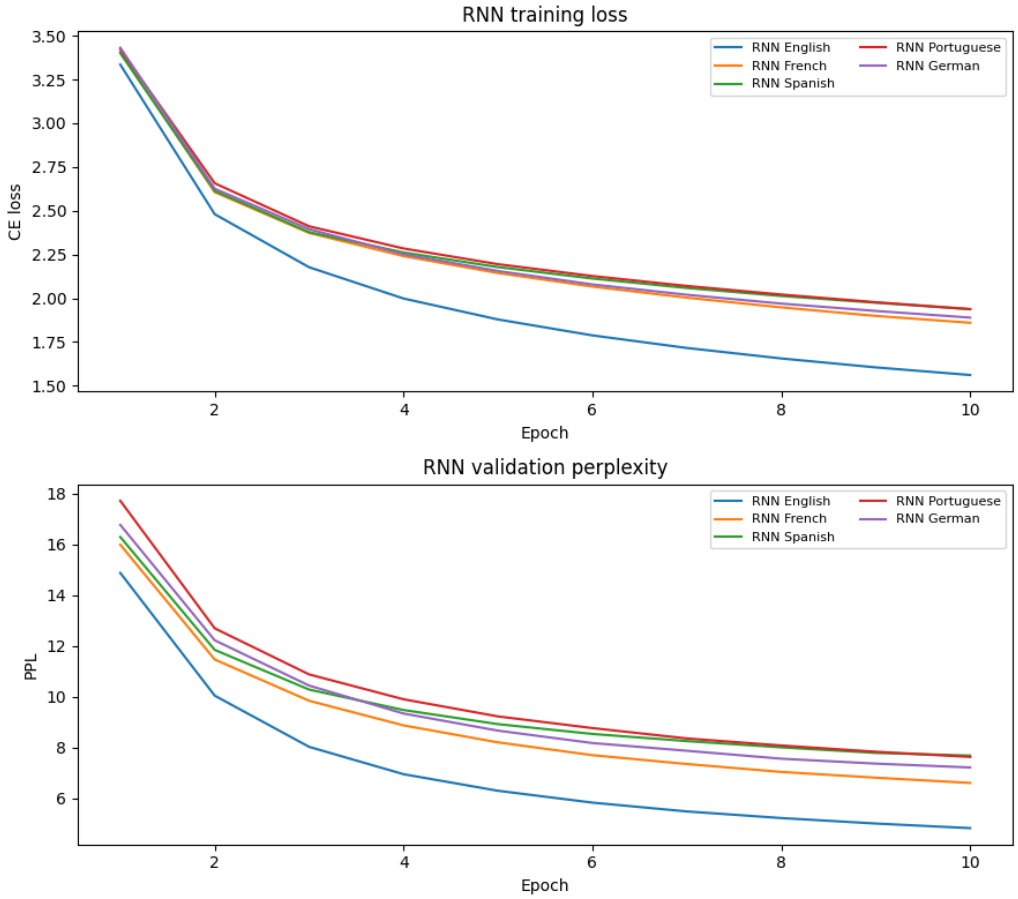


Figure 2: RNN training loss (top) and validation perplexity (bottom) for the five languages.

As Figure 2 shows, the training loss decreases steadily for all languages, falling from about 3.4 to between 1.56 and 1.94. The validation perplexity follows the same trend. Both curves are still decreasing at epoch 10, so the last epoch is selected for every language and no overfitting is observed. The best per-language validation perplexities are:

- English: 4.83
- French: 6.60
- German: 7.21
- Portuguese: 7.63
- Spanish: 7.68

English is the easiest to model because its spelling uses almost no diacritics.

Table 2: Test confusion matrix of the RNN models (rows: true language, columns: predicted language).

True \ Pred	English	French	Spanish	Portuguese	German
English	139	1	0	0	0
French	0	140	0	0	0
Spanish	0	1	136	3	0
Portuguese	0	2	2	135	1
German	0	0	0	0	140

Results. On `test.csv` the RNN models obtain:

- Accuracy: 0.9857
- Macro-F1: 0.9857
- Corpus-level test perplexity: 5.7054 (validation: 5.8937)

Cost. Training the five models takes 6.4 s in total on the GPU, with a peak GPU memory of 391.7 MB. Inference over the test set takes 4.1 s because every text must be passed through all five models.

Correct and incorrect predictions. Even very short texts are identified correctly when they contain language-specific function words or diacritics. Examples include "*Il est si*" (French, 9 characters), "*é preciosa,*" (Portuguese), and "*Y escalamos*" and "*Son una red*" (Spanish). As Table 2 shows, the 10 errors are concentrated on Spanish and Portuguese. These are short phrases made of words that are spelled the same or almost the same in both languages, which leaves too little evidence to separate them.

1.4 Character-Level LSTM Language Models

The LSTM models use exactly the same split, vocabulary, loss, optimiser, and hyperparameters as the RNN models. The only change is that the RNN layers are replaced with LSTM layers.

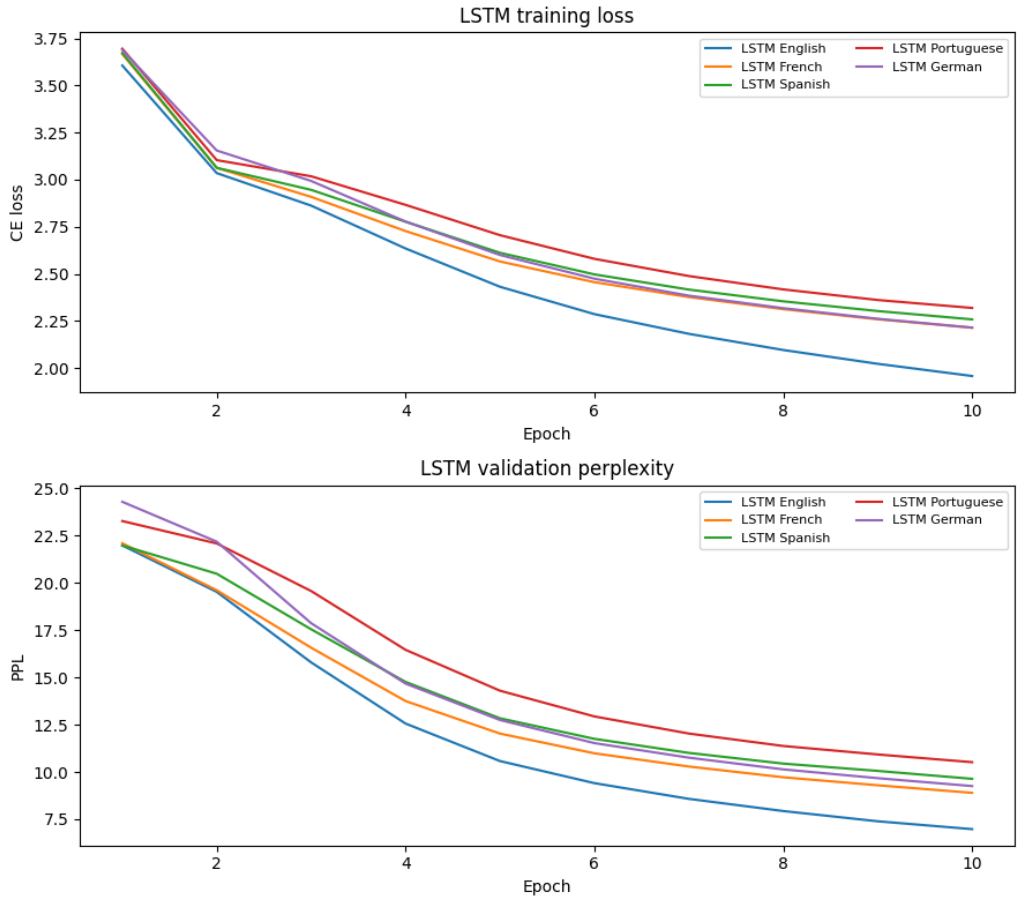


Figure 3: LSTM training loss (top) and validation perplexity (bottom) for the five languages.

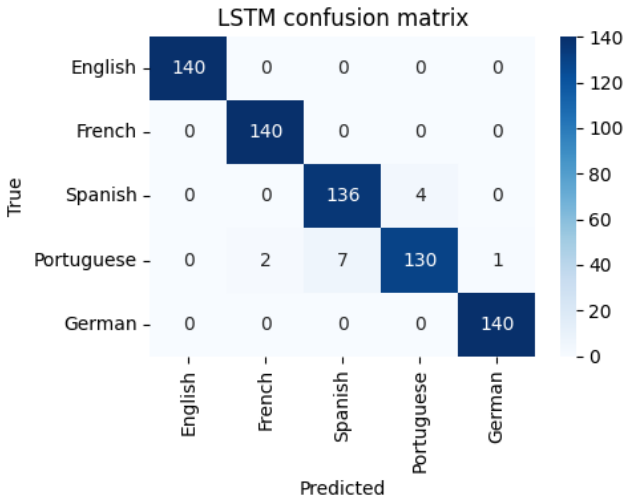


Figure 4: Test confusion matrix of the LSTM models.

Learning curves. The LSTM curves (Figure 3) also decrease monotonically, but they start higher and converge more slowly than the RNN curves. After 10 epochs, the training loss is between 1.96 and 2.32, and the best validation perplexities are:

- English: 6.98
- French: 8.90
- German: 9.25
- Spanish: 9.64
- Portuguese: 10.52

These are all higher than the RNN values. An LSTM has four gates per step and therefore about four times as many recurrent parameters, so the same 10-epoch budget leaves it further from convergence.

Results. On `test.csv` the LSTM models obtain:

- Accuracy: 0.9800
- Macro-F1: 0.9799
- Corpus-level test perplexity: 7.9834 (validation: 8.1334)

Cost. Training takes 9.0 s, with a peak GPU memory of 563.3 MB, and inference over the test set takes 5.1 s. Both are higher than for the RNN because of the larger recurrent cell.

Correct and incorrect predictions. Short texts such as "*Il est si*", "*Se comunican*", and "*as intuições*" are identified correctly. As Figure 4 shows, all 14 errors are Spanish or Portuguese texts: 7 Portuguese texts are predicted as Spanish, 4 Spanish texts as Portuguese, and the remaining 3 Portuguese texts as French or German.

1.5 Comparison and Discussion

Table 3: Comparison of the four model families on `test.csv`.

Model	Val PPL	Test PPL	Accuracy	Macro-F1	Train (s)	Inference (s)	Peak memory
Bigram	11.44	11.33	0.9729	0.9727	1.91	0.17	0.42 MB (CPU)
Trigram	9.70	9.40	**0.9871**	**0.9871**	2.12	0.18	2.64 MB (CPU)
RNN	**5.89**	**5.71**	0.9857	0.9857	6.42	4.08	391.7 MB (GPU)
LSTM	8.13	7.98	0.9800	0.9799	9.01	5.13	563.3 MB (GPU)

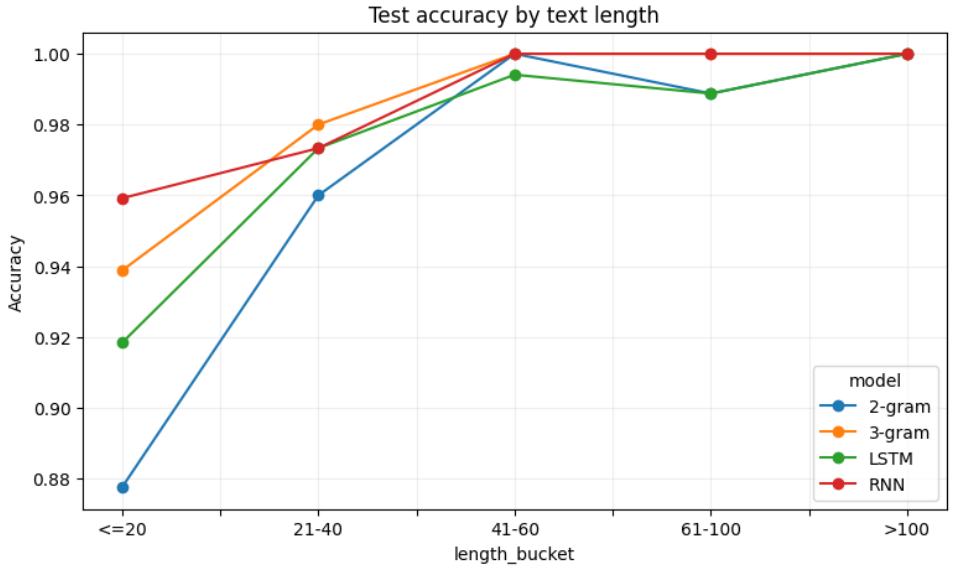


Figure 5: Test accuracy as a function of text length (number of characters).

Hardware and computation time. The n-gram models only need a CPU and a few megabytes of memory. They are built in about 2 s and classify the entire test set in under 0.2 s. The RNN and LSTM models need a GPU, with 392 MB and 563 MB of peak GPU memory respectively. They take longer to train, and their inference is more than 20 times slower, because every text requires a sequential forward pass through five neural models.

Does lower perplexity always mean higher accuracy? No. The RNN has by far the lowest perplexity (5.71, compared with 9.40 for the trigram), yet the trigram achieves the highest accuracy (0.9871 versus 0.9857). Likewise, the LSTM has lower perplexity than the trigram but lower accuracy. Corpus perplexity measures how well the correct language model predicts its own texts, averaged over all characters. Identification, however, depends only on the relative ranking of the five models for each individual text. A model can model long texts very well and still rank the wrong language first on a few short, ambiguous texts. Lower perplexity therefore tends to help, but it does not guarantee better identification accuracy.

Most frequently confused languages. Summed over all four models, the most frequent directed confusions are:

- Portuguese → Spanish: 19
- Spanish → Portuguese: 14
- Portuguese → French: 7
- Portuguese → German: 6

Spanish and Portuguese are by far the most confusable pair, because they share most of their vocabulary and character patterns. English, French, and German are almost never confused with one another.

Are shorter texts harder? Yes. As Figure 5 shows, every model performs worst on the shortest texts (at most 20 characters, 49 test samples). In that group, accuracy is 0.878 for the bigram, 0.918 for the LSTM, 0.939 for the trigram, and 0.959 for the RNN. Accuracy then rises with length:

- 21–40 characters: 0.960–0.980
- 41 characters or more: 0.989–1.000

Short texts provide only a few characters of evidence, and a single number, name, or shared word can dominate the score. Longer texts accumulate enough language-specific patterns to be classified almost perfectly. The RNN is the most robust model on short texts, while the bigram is the least robust.

2 Self-Attention Layer for Semantic Similarity Scoring

2.1 Implementation Rules and Model Design

The provided notebook `atten.ipynb` was used as the starting point. Following the course announcement, only the code inside the `TODO` and `TODO (Optional)` blocks was changed. The data loading, vocabulary, dataset, training loop, evaluation, and submission code are the provided versions. Additional cells were appended only to run the experiments in (c) and (d) and the ensemble in (e). These cells call the provided functions and do not modify them. The ensemble is placed in a separate code cell, as permitted by the TA.

Each example is converted into a single token sequence

$$z = \texttt{<context>}\ \text{context}\ \texttt{<sep>}\ \text{anchor}\ \texttt{<sep>}\ \text{target},$$

using the provided tokenizer and the vocabulary built from `train.csv` (vocabulary size 7,749). The fixed settings are `MAX_LEN` $= 64$, $d_{\text{model}} = 128$, 2 encoder layers, batch size 64, seed 42, and the MSE training loss. Model selection uses the epoch with the highest validation Pearson correlation, which the provided `train_model` already does.

Multi-head self-attention (TODO). The layer is implemented from scratch, without PyTorch's built-in multi-head attention module (`torch.nn.MultiheadAttention`). Q, K, and V are linear projections of the input, split into h heads of size $d_k = 128/h$. For each head,

$$\text{Attn}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}} + M\right) V,$$

where M sets the scores of padded key positions to -10^9. The heads are concatenated and passed through an output projection.

Encoder layer (TODO). Each layer uses post-layer normalisation as in the original Transformer,

$$x \leftarrow \text{LN}\big(x + \text{Drop}(\text{MHA}(x))\big), \qquad x \leftarrow \text{LN}\big(x + \text{Drop}(\text{FFN}(x))\big),$$

with FFN = Linear $\rightarrow$ ReLU $\rightarrow$ Dropout $\rightarrow$ Linear. In the tuned model, the layer input is first normalised by an extra LayerNorm. The provided embedding is initialised from $\mathcal{N}(0, 1)$ and multiplied by $\sqrt{128} \approx 11.3$, while the attention output is of order 1. Without the input normalisation, $x + \text{MHA}(x)$ in the first layer is dominated by the embedding and the first attention sub-layer has almost no effect.

Pooling (TODO Optional). The anchor and target positions are recovered from `input_ids` alone: cumsum(`input_ids` = `<sep>`) is 0 for the context, 1 for the anchor, and 2 for the target. From the encoder output, masked means are taken over the whole sequence (g), the anchor words (u), and the target words (v). Three strategies were used:

- **mean** (default): masked mean over all tokens.
- **interact** (parameter-free): a 128-dimensional vector made of three slices, g (32 dimensions), $u \odot v$ (48), and $|u - v|$ (48).
- **mlp**: the full vector $[g, u, v, |u-v|, u \odot v]$ is passed through Linear($5d \to d$) $\to$ GELU $\to$ Dropout. Optional extra inputs are token-level soft-alignment features (ESIM-style; "am", "adf") and max-pooled anchor/target vectors ("umax", "vmax"). The projection is created inside the pooling TODO block.

In every case the output has size d, so the provided regression head $\hat{s} = \sigma(W h_{\text{pool}} + b)$ is reused unchanged.

Optimiser and scheduler (TODO). AdamW is used with two parameter groups. The token embedding has its own learning rate (a multiple of the base rate), because its weights are of order 1 and an Adam step of 10^{-3} changes them by only about 0.1%. The scheduler is a per-epoch linear warm-up (2 epochs) followed by cosine decay.

Final setting. 4 heads, `mlp` pooling with $[g, u, v, |u - v|, u \odot v]$, input LayerNorm, dropout 0.1, base learning rate 10^{-3} with a 20$\times$ embedding learning rate, weight decay 0.01, warm-up + cosine schedule, and 25 epochs.

2.2 (a) Learning Curves with 4 Heads

Figure 6 shows the learning curves of the main 4-head model (final setting).

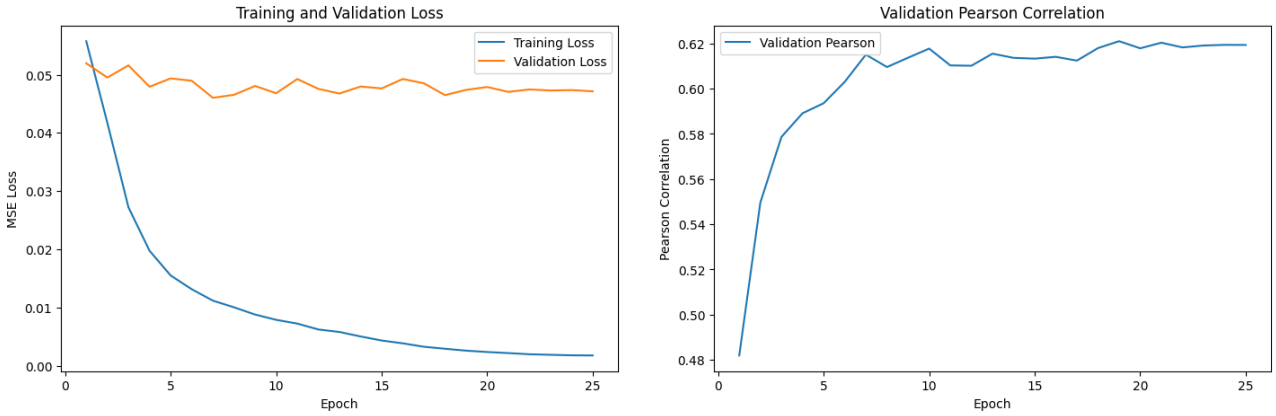


Figure 6: Main model with 4 heads. Left: training and validation MSE loss. Right: validation Pearson correlation.

The training loss falls quickly, from 0.0558 at epoch 1 to 0.0018 at epoch 25. The validation loss reaches its minimum of 0.0460 at epoch 7 and then stays between 0.046 and 0.049. After a few epochs the model therefore fits the training set far better than the validation set, which is clear overfitting. The validation Pearson correlation still rises after the validation loss stops improving: from 0.4819 at epoch 1 to 0.6151 at epoch 7, reaching its maximum of 0.6210 at epoch 19. The model keeps improving the ranking of the pairs even when the squared error no longer decreases. Because the provided `train_model` keeps the checkpoint with the best validation Pearson, the late-epoch overfitting does not harm the selected model.

2.3 (b) Validation Performance

Using the checkpoint with the best validation Pearson correlation (epoch 19), the 4-head main model obtains

$$\text{Pearson} = 0.6210, \qquad \text{RMSE} = 0.2177, \qquad \text{MSE} = 0.0474$$

on `val.csv`. For reference, the default baseline with 4 heads (mean pooling, no scheduler, 10 epochs; D0 in Table 5) reaches Pearson 0.3173 and RMSE 0.2487.

2.4 (c) Effect of the Number of Attention Heads

The number of heads was set to 1, 4, and 8. The data split, seed, optimiser, and 10-epoch budget were kept the same, using the default baseline setting (mean pooling, AdamW with learning rate 10^{-3} and weight decay 10^{-4}, no scheduler).

Table 4: Validation performance for different numbers of attention heads.

Heads	Head dimension	Best val Pearson	Val RMSE	Training time (s)
1	128	**0.3282**	**0.2474**	74.6
4	32	0.3173	0.2487	76.2
8	16	0.2968	0.2491	76.1

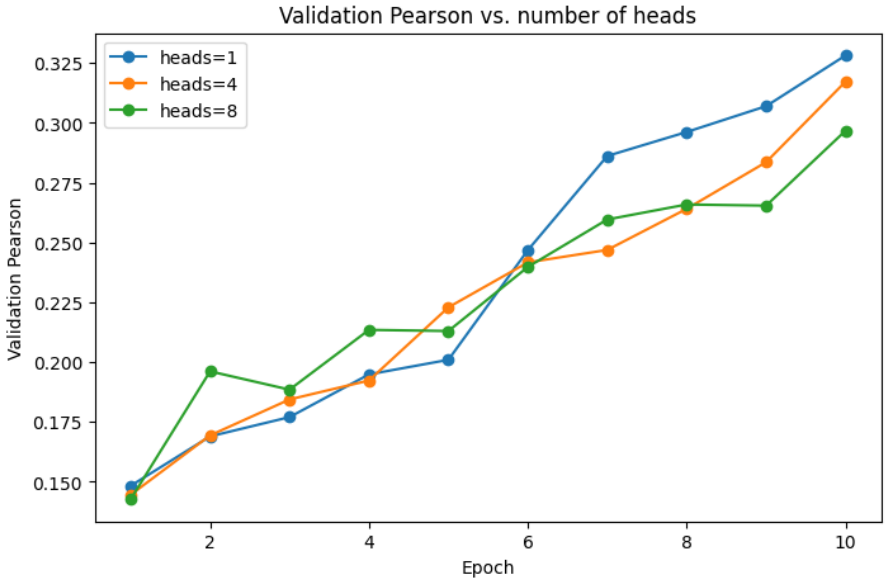


Figure 7: Validation Pearson correlation per epoch for 1, 4, and 8 attention heads.

All three settings have the same number of parameters, because the Q, K, V, and output projections are always 128×128. Only the way the 128-dimensional space is split into heads changes.

As Table 4 shows, fewer heads perform slightly better: 1 head reaches 0.3282, 4 heads 0.3173, and 8 heads 0.2968. The RMSE values are almost identical (0.2474 to 0.2491). The inputs are very short (about 8 tokens), and the main signal is a word-level match between the anchor and the target. A single head with 128-dimensional queries and keys has a richer similarity function. With 8 heads, each head computes its attention scores in only 16 dimensions, which makes them noisier. Training time is nearly the same for all three (74.6–76.2 s).

The differences are small (0.03 Pearson at most), and all curves are still rising at epoch 10 (Figure 7). The comparison therefore mainly reflects how fast each setting converges under the same budget. The number of heads is not the bottleneck for this task. Much larger gains come from the changes in (d), so the 4-head setting required by the assignment is kept for the main model.

2.5 (d) Tuning Permitted Settings

Starting from the baseline D0, each configuration adds one documented change, using only settings inside the `TODO` / `TODO (Optional)` blocks: hyperparameters, the optimiser, the scheduler, the pooling strategy, and the inside of the attention and encoder-layer classes. Every configuration is trained from seed 42.

Table 5: Tuning experiments on `val.csv`. "Base" is the configuration the change is applied to.

ID	Base	Change	Val Pearson	Val RMSE	Best epoch
D0	–	Baseline: mean pooling, AdamW (wd 10^{-4}), no scheduler, 10 epochs	0.3173	0.2487	10
D1	D0	25 epochs, 2-epoch warm-up + cosine decay, weight decay 0.01	0.4196	0.2436	25
D2	D1	`interact` pooling (parameter-free)	0.4871	0.2568	24
D3	D2	Input LayerNorm in each encoder layer	0.5175	0.2540	25
D4	D3	Relative-position bias in attention	0.5260	0.2505	25
D5	D3	10× embedding learning rate	0.5919	0.2267	22
D6	D3	30× embedding learning rate	0.5965	0.2249	25
D7	D5	Relative-position bias	0.6001	0.2249	25
D8	D5	`mlp` pooling $[g, u, v, \lvert u - v\rvert, u \odot v]$	0.6161	0.2310	21
D9	D8	Dropout $0.2 \to 0.1$	0.6262	0.2143	20
D10	D9	20× embedding learning rate (**final setting**)	0.6394	0.2108	14
D11	D10	+ alignment features (am, adf)	0.6244	0.2103	23
D12	D10	+ max-pooled features (umax, vmax)	**0.6400**	**0.2091**	16
D13	D10	+ alignment and max-pooled features	0.6324	0.2093	16

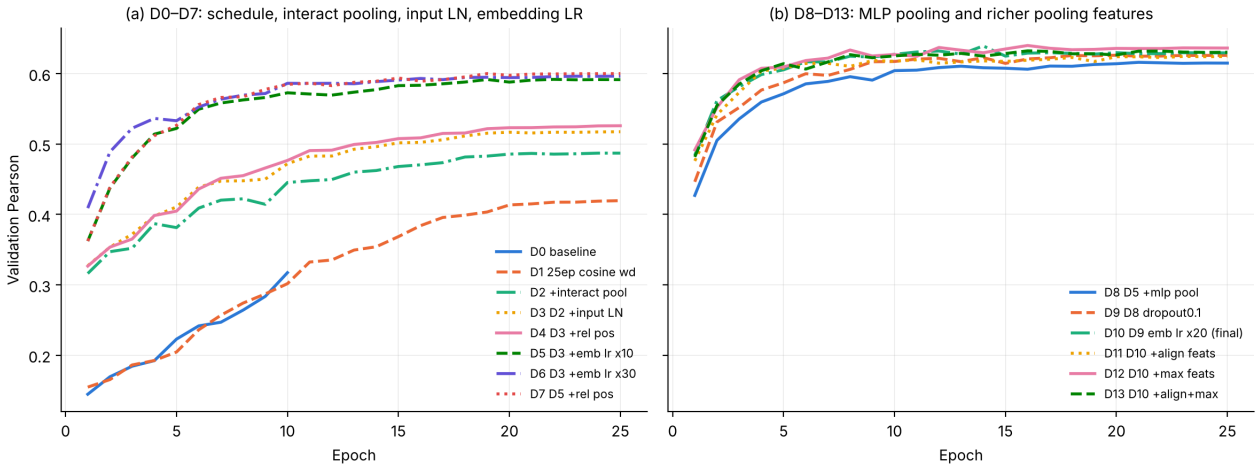


Figure 8: Validation Pearson correlation per epoch for the tuning configurations. (a) D0–D7. (b) D8–D13.

D1: longer budget, schedule, and weight decay (+0.102). The baseline is still improving at epoch 10. With 25 epochs, a warm-up + cosine schedule, and stronger weight decay, the validation Pearson rises to 0.4196. Figure 8a shows that the D1 curve is still rising at epoch 25, so mean pooling learns slowly.

D2: interaction pooling (+0.068). Mean pooling mixes the context, anchor, and target into one vector, so the regression head cannot tell which phrase a feature comes from. The parameter-free `interact` pooling exposes the alignment $u \odot v$ and the difference $|u - v|$ directly and raises the Pearson correlation to 0.4871. Its RMSE is slightly higher (0.2568), probably because the fixed slices change the scale of the features fed to the linear head. Model selection uses Pearson correlation, so this is acceptable.

D3–D7: fixing the embedding scale. The input LayerNorm (D3) lets the first attention sub-layer contribute and adds 0.030. The relative-position bias adds little: +0.009 on D3 (D4) and +0.008 on D5 (D7). The largest gain in this group comes from the embedding learning rate. A 10× rate (D5) raises the Pearson correlation from 0.5175 to 0.5919 (+0.074) and lowers the RMSE from 0.2540 to 0.2267. The embedding table holds most of the model's parameters, and with the base rate it hardly moves from its random initialisation. A 30× rate (D6) gives only a further +0.005, and the bias was not kept because of its small effect.

D8–D10: MLP pooling, dropout, and embedding learning rate. Replacing the fixed slices by a learnable projection of the full interaction vector (D8) adds 0.024, since the model can now combine g, u, v, $|u-v|$, and $u \odot v$ non-linearly. With the learnable pooling, dropout 0.2 is too strong. Lowering it to 0.1 (D9) adds 0.010, and re-tuning the embedding learning rate to 20× (D10) adds another 0.013, reaching 0.6394. D10 is the final setting of the main model. As Figure 8b shows, these configurations reach their best Pearson correlation within 14–21 epochs.

D11–D13: richer pooling features. Max-pooled anchor and target vectors (D12) reach 0.6400 and the lowest RMSE (0.2091), almost the same as D10 (+0.0006). The alignment features alone (D11) lower the Pearson correlation by 0.015, and using both (D13) gives 0.6324. A likely reason is that self-attention already lets every anchor token attend to the target tokens, so explicit soft alignment adds little new information. Because D10, D12, and D13 are within 0.01 of the best Pearson correlation, all three feature sets are used as members of the final ensemble.

Run-to-run variation. The main model in (a) uses the same setting as D10 but reaches 0.6210 instead of 0.6394. The only difference is the state of the random-number stream (initialisation, data order, and dropout). Differences of about 0.02 between runs are therefore within the noise of a single run. This variance is the motivation for the ensemble in (e).

2.6 (e) Final Model and Kaggle Submission

The final model is an ensemble of eight models, implemented in an extra code cell as permitted by the TA. Member 0 is the main model from (a). The other seven members are trained with the provided `TransformerRegressor` and `train_model` and the same optimiser and scheduler. To make the members' errors less correlated, they cycle through the three pooling feature sets selected in (d): $[g, u, v, |u-v|, u \odot v]$ (D10), the same plus max-pooled features (D12), and the same plus alignment and max-pooled features (D13). The seed is not reset between members, so every member continues the seed-42 random stream and receives a different initialisation, data order, and dropout pattern. The members' sigmoid outputs are averaged.

Table 6: Validation results of ensemble members 0–5 (best-Pearson checkpoint of each member).

Member	Pooling features	Val Pearson	Val RMSE	Best epoch
0	$[g, u, v, \lvert u-v\rvert, u \odot v]$ (main model)	0.6210	0.2177	19
1	+ umax, vmax	**0.6426**	0.2081	16
2	+ am, adf, umax, vmax	0.6368	**0.2063**	9
3	$[g, u, v, \lvert u-v\rvert, u \odot v]$	0.6322	0.2153	19
4	+ umax, vmax	0.6298	0.2134	15
5	+ am, adf, umax, vmax	0.6370	0.2078	13

Table 7: Summary of the main models on `val.csv`.

Model	Val Pearson	Val RMSE
Baseline (D0: 4 heads, mean pooling, 10 epochs)	0.3173	0.2487
Main model (a)/(b) (final setting, 4 heads)	0.6210	0.2177
Best single configuration in (d) (D12)	0.6400	0.2091

The members reach validation Pearson correlations between 0.6210 and 0.6426, a spread of about 0.02 for essentially the same setting. This spread is the run-to-run variance that averaging removes: the members make partly independent errors, so the averaged prediction is more stable than any single model. Overall, the validation Pearson correlation improves from 0.3173 for the baseline to 0.6210–0.6426 for the tuned single models, and the averaged ensemble is used for the final submission.

The final ensemble was used to predict all 7,295 samples in `test.csv`. The IDs are taken from `test.csv` in their original order, and the same assertions as in the provided submission cell check the row count and the ID order. This submission obtained a Kaggle leaderboard score of **0.68679** (Figure 9).

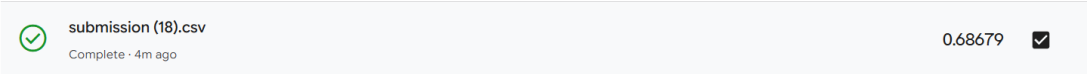


Figure 9: Kaggle leaderboard score of the final submission (0.68679).